



TRAINING BY BCN DRONE CENTER HOSTED BY BIRDVIEW MAPPING

DESIGN AND CONSTRUCTION OF DRONE



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SCOPE

This document describes both the Technical and Economical Proposal for performing a custom training focused on drone design and construction. During the training, participants will be mounting a full drone system in small groups, which will be functional by the end of the training.

The main training will be also completed with sessions about image processing, drone applications, maintenance and advanced drone pilot & operation.



TECHNICAL PROPOSAL

The training will be executed in India at client office, and with a duration of 4 weeks.

The proposed planning will be:

Week 1 (Monday to Friday): First training week, focused on all basic knowledge, with the initial steps of building the system.

Week 2: Second training week, with a strong focus on hand-on training with the fulfilment of the construction of the drones, their operational set-up and testing. By the end of the week all systems would be ready and flight tested.

Week 3 (Only in 3 weeks training option): Third training week, with a focus on advanced systems, payload integration and advanced operational procedures.

Week 4 (Only in 4 weeks training option): Fourth training week, with a focus on piloting different systems, included the ones constructed during the training. Dates can be re-arranged at convenience until 45 days before the training starts.



DRONE CONSTRUCTION

The course has a strong focus in hands-on training. Therefore, during the training 2 different drones are going to be built by the group:

One VTOL system, with autonomous operation capability and the capacity to carry at least one RGB camera.

One multirotor system, with autonomous operation capability and the capacity to carry at least one Lidar Sensor / RGB camera.

As part of this training process, there will be 2 tested and ready to fly drones that the company contracting the course can keep for his own use.

DRONE CONSTRUCTION

One VTOL system, with autonomous operation capability and the capacity to carry at least one RGB camera.
200km Radio network tested in harsh terrain in Barcelona for transport of medical supplies payloads available



DRONE CONSTRUCTION

One multicopter system, with autonomous operation capability and the capacity to carry at least one Lidar Sensor / RGB camera.



TRAINING SCHEDULE

The training will be divided in 3 different modules:

Module 1 – Design & Construcion Training.

Module 2 – Piloting & Operational Procedures Training.

Module 3 – Data Processing, Sensors & Applications Training.

The training will be performed from 9:00 to 13:00 and 15:00 to 17:00 every day from Monday to Friday. A detailed schedule with what will be done in every hour will be presented once the offer has been accepted.

TRAINING SYLLABUS

The extension of the syllabus will depend on picking the 2 weeks, 3 weeks or 4 weeks options and on the preferable options of the purchasing company. Here you can find a list of all main topics that can be covered inside each Module:

MODULE 1 – DESIGN & CONSTRUCTION

- Aerodynamics
- Stability & Control
- Performance
- Propulsion
- Navigation
- On-board systems: Sensors
- On-board systems: Actuators
- On-board systems: Automatic Pilots
- On-board systems: Communication Systems
- On-board systems: Communication Systems
- On-board systems: Advanced Systems
- Ground Control Stations
- Fixed-wing design principles
- Multicopter design principles
- VTOL design principles
- Helicopter design principles

MODULE 2 — PILOTING & OPERATIONAL PROCEDURES

- Flight Planning
- Operational Procedures
- Meteorology
- Human Factors
- Multicopter Piloting & Operation
- Fixed-wing Piloting & Operation
- VTOL Piloting & Operation
- Manual piloting
- Autonomous flight planning
- Autonomous operation
- VLOS operation
- EVLOS operation
- BVLOS operation
- Drone Legislation
- Radio Communications
- Airspace Classification

BCN CENTER TRAINING STAFF

Training Staff

Jordi Santacana Pérez

CEO and Founder at BCN Drone Center & CATUAV

**Education:**

Pilot & Flight Instructor

Experience:

With more than 30 years experience in the development and operation of UAS, Jordi Santacana has been one of the pioneers of drone civil technology.

Training Tasks:

He will be performing drone design lessons and piloting training, taking advantage of the knowledge gathered after having design more than 40 different drones.

<https://www.linkedin.com/in/jordi-santacana-84a83b102/>

Jordi Salvador Bernadí

CTO and Business Development at BCN Drone Center

**Education:**

Master of Science in Engineering (M.Sc.Eng.), Computer Vision and Artificial Intelligence (UAB)

Master of Science in Engineering (M.Sc.Eng.), Aerospace Engineering (UPC)

Company Creation & Administration (EADA)

Experience:

Jordi Salvador has more than 15 years experience in the development of aerial robotic applications, combining the knowledge of aerospace engineering with imagery processing, computer vision and Artificial Intelligence.

Training Tasks:

He will be performing module 2 & 3 sessions, bringing his large experience with drone systems and its applications.

<https://www.linkedin.com/in/uavengineer/>

BCN CENTER TRAINING STAFF

Lluís Amat Olóndiz

UAS Regulatory Compliance at BCN Drone Center

**Education:**

Master of Science in Engineering (M.Sc.Eng.), Aeronautical Engineering, 2014-2016, UPC - Universitat Politècnica de Catalunya Bachelor Degree of Science in Engineering (B.Sc.Eng.), Aerospace Engineering, 2010-2014, UPC - Universitat Politècnica de Catalunya.

Experience:

Having worked for both the Spanish National Aviation Authority (NAA) and the private UAS sector, Lluís possesses a deep understanding of the AAM regulation framework from both the regulator and industry perspectives.

Training Tasks:

He will be performing all regulation sessions.

<https://www.linkedin.com/in/lluisamatolondriz/>

Alfredo Gil

Head of UAV Design & Production at CATUAV

**Education:**

Automotive Construction Tech Engineer

Experience:

Alfredo has more than 30 years' experience working in the design and construction of automotive prototypes and products. He has brought it's large experience in the automotive sector to the drone industry, in which he has been working for the past 10 years.

Training Tasks:

He will be performing most of the construction training.

<https://www.linkedin.com/in/alfredo-gil-59604045/>

BCN CENTER TRAINING STAFF

Eduard Miralles Llauradó

R+D Engineer in CATUAV

**Education:**

Master's Degree in Applications and Technologies for Unmanned Aircraft Systems (UPC).

Master of Science in Engineering (M.Sc.Eng.), Aerospace Engineering (UPC).

Bachelor's Degree in Engineering, Aerospace Engineering (UPC).

Experience:

Eduard has a long extensive working with multiple robotic systems, both in the ground or aerial. He is charge of technical R+D development of UAS new solutions in CATUAV.

Training Tasks:

He will be performing all sessions and practical training related with avionics, autopilots, software and electronics integrations.

<https://www.linkedin.com/in/eduardmiralles/>

Pol Ayats Plana

Operations Engineer in BCN Drone Center

**Education:**

Engineer's degree, Aerospace Systems (UPC)

Experience:

As BCN Drone Center Operations Engineer, Pol Ayats has an extensive experience both in the legal and technical parts of UAS operation, being able to properly plan any type of mission.

Training Tasks:

He will be performing drone planning and operation lessons, both theoretical and practical, bringing his experience operating multiple types of UAS.

<https://www.linkedin.com/in/polayats/>

QUOTATION

Custom Design & Construction Training (8 to 10) people , 4 week training

Cost of 2 Drone design and construction cost for below mentioned Drones :

Total in rupee : (Rs 8,50,00,000 /-) Eight crore fifty lakh

One VTOL system, with autonomous operation capability and the capacity to carry at least one RGB camera for defence.

One multirotor system, with autonomous operation capability and the capacity to carry at least one Lidar Sensor / RGB camera.

PAYMENT CONDITIONS

100% payment on confirmation of the training program on PO issue

COMPANY PRESENTATION



BCN Drone Center is a UAV Test Site for civil air systems. Inaugurated in 2014, it has had users from 100+ different nationalities, being an international benchmark in UAS sector.

The center has a segregated airspace of 5.000 hectares with a flying altitude of up to 4.500 feet AMSL, enabling the performance of experimental flights Beyond Visual Line of Sight (BVLOS) or at higher altitudes, being much easier to get flight approvals for any type of drone, even those above 25 kg or 3 meters wingspan.

The Test Site has all necessary infrastructure to carry out any R+D testing program, including an hangar, ground control station, workshop, electronics lab, office space... The facilities are placed in the middle of an outdoor laboratory, set in the 14 ha of proprietary terrain, where in addition of the two runways the users can find multiple targets to calibrate and validate their sensors and experiments. Furthermore, there is plenty of room to install any required elements for any testing or flight demo.

BCN Drone Center is managed by CATUAV, a pioneer company in Europe in the civil drone sector (founded in year 2000), sharing its wide experience in the sector to give consultancy services to all Test Site users when needed.

The combination of our long experience with our unique testing facilities, allows us to offer one-stop shop services to deploy to market any UAS project or idea, from its conception to its final commercial implementation.

This includes feasibility & viability studies, engineering and R+D projects, tech + legal + market consultancy, operational authorisation processing (SORA, Design Verification...), professional training and the full testing + evaluation + certification pipeline.





THANK YOU

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